Hall Ticket Number:

MR-21

Code: B313CS5S

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. III Semester Supplementary Examinations, Aug/Sep-2023

Discrete Mathematics

(Common to CSE, CSM & CSD)

Time: 3 Hours | *Answer ALL Questions* | Max. Marks: 70

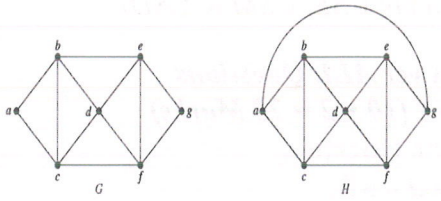
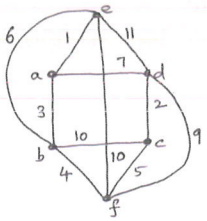
Part-A (10 x 2 = 20 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	Construct the truth table for $\neg(P \vee (Q \wedge R))$.	2	2	1	1,2
1. b)	Test the validity of the following argument $(p \wedge r) \rightarrow (\neg q)$, $(\neg q) \rightarrow r$ $\therefore (p \wedge r) \rightarrow r$	2	3	1	1,2
1. c)	What is partial ordering?	2	1	2	1
1. d)	Define cardinality of sets with an example.	2	1	2	1
1. e)	Write the statement of pigeonhole principle.	2	3	3	1
1. f)	Find out how many 5-digit numbers greater than 30,000 can be formed from the digits 1,2,3,4 and 5.	2	2	3	1
1. g)	What is generating function?	2	1	4	1
1. h)	Define axioms of probability.	2	1	4	1
1. i)	Define graph coloring with an example.	2	1	5	1
1. j)	What is spanning tree?	2	1	5	1

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	Show that $(P \rightarrow Q) \rightarrow R$ and $P \rightarrow (Q \rightarrow R)$ are logically equivalent	5	3	1	1,2
2. b)	Show the following equivalence without constructing the truth table. $((P \wedge Q \wedge A) \rightarrow C) \wedge (A \rightarrow (P \vee Q \vee C)) \Leftrightarrow (A \wedge (P \leftrightarrow Q)) \rightarrow C$	5	3	1	1,2
	OR				
2. c)	State and explain the rules that can generate a well formed formula.	5	2	1	1
2. d)	Obtain the principal disjunctive and conjunctive normal form of the following formula. $(P \rightarrow (Q \wedge R)) \wedge (\neg P \rightarrow (\neg Q \wedge \neg R))$	5	3	1	1,2
3. a)	Write the properties of binary relations.	5	3	2	1
3. b)	Find the transitive closure of the relation $R = \{(1, 2), (2, 3), (3, 4), (4, 1)\}$. Show that R^i for all values i for that give new elements of the transitive closure.	5	2	2	1,2
	OR				
3. c)	What is n-ary relations? And explain its applications.	5	1	2	1
3. d)	Let A= {1, 2, 3, 4, 6, 8, 12, 24}, show that the relation "divides" is a partial ordering on A and draw the Hasse diagram.	5	3	2	1,2
4. a)	How many four letter words can be formed using the letters $a, a, a, b, b, c, c, c, c, d, d$?	5	2	3	1
4. b)	Expand $(2x - y)^7$ using the Binomial Theorem.	5	3	3	1
	OR				
4. c)	Use mathematical induction to show that $1 + 2 + 2^2 + \ldots\ldots\ldots\ldots + 2^n = 2^{n+1} - 1$ for all positive integers n.	5	3	3	1
4. d)	Write the 3-combinations and 3-permutations of **{3.a, 2.b, 1.c, 3.d}**.	5	3	3	1
5. a)	Solve the recurrence relation $a_n - 7a_{n-1} + 10a_{n-2} = 4^n$, $a_0=0$, $a_1=1$.	5	3	4	1,2
5. b)	Find the coefficient of x^{2005} in the generating function $1/(1+5x)^2$	5	2	4	1

P.T.O.

	OR				
5. c)	Solve the recurrence relation $a_n = 2a_{n-1} + 3a_{n-2}$ for $n \geq 2$ where $a_0 = 2$ and $a_1 = 2$.	5	3	4	1,2
5. d)	State and Prove Baye's Theorem.	5	2	4	1,2
6. a)	What are the chromatic numbers of the graph *G* and *H*? 	5	1	5	1
6. b)	Write down the applications of trees.	5	2	5	1
	OR				
6. c)	State and Prove fundamental theorem of graph theory.	5	2	5	1,2
6. d)	Find the minimal spanning tree using Kruskal's algorithm. 	5	2	5	1

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome